

If we look back at our derivation of the heat equation, we can make things more general by:

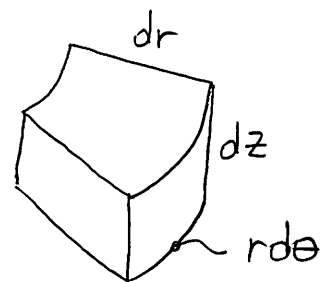
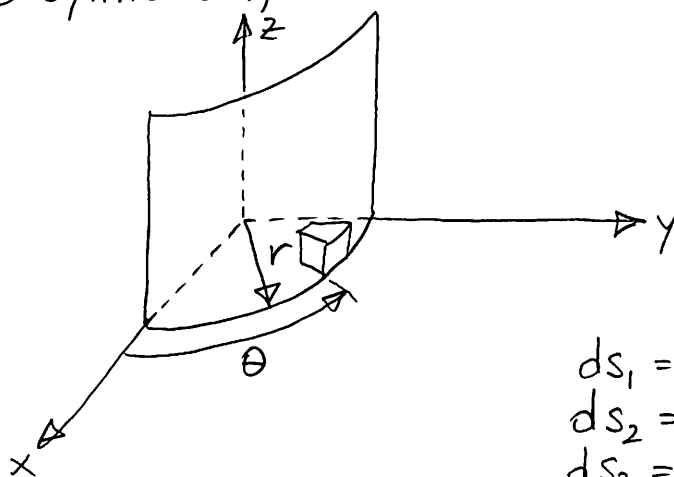
$dV = ds_1 ds_2 ds_3$ where s_1, s_2, s_3 are the coordinates in consideration

$$\frac{1}{ds_1 ds_2 ds_3} \left\{ \frac{\partial}{\partial s_1} \left(k \frac{\partial T}{\partial s_1} \right) ds_1 + \frac{\partial}{\partial s_2} \left(k \frac{\partial T}{\partial s_2} \right) ds_2 + \frac{\partial}{\partial s_3} \left(k \frac{\partial T}{\partial s_3} \right) ds_3 \right\} + Q''' = \rho c_p \frac{\partial T}{\partial t}$$

Specific Cases:

① Cartesian, $\left. \begin{array}{l} ds_1 = dx \\ ds_2 = dy \\ ds_3 = dz \end{array} \right\}$ We'll get what we already solved

② Cylindrical,



$$\begin{array}{l} ds_1 = dr \\ ds_2 = dz \\ ds_3 = r d\theta \end{array}$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(k r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q''' = \rho c_p \frac{\partial T}{\partial t}$$

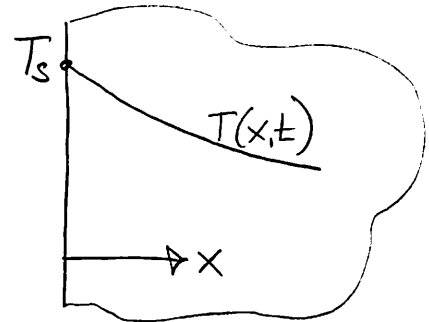
↳ Heat equation in cylindrical coordinates.

Boundary Conditions

① Dirichlet (1'st kind)

$T = \text{specified on the boundary}$

$$T(0, t) = T_s \quad \text{or}$$

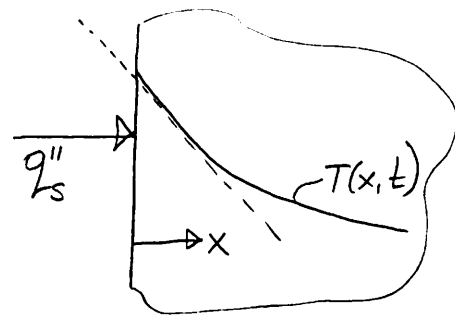


② Neumann (2'nd kind) or Constant surface heat flux

(a) Finite Heat Flux

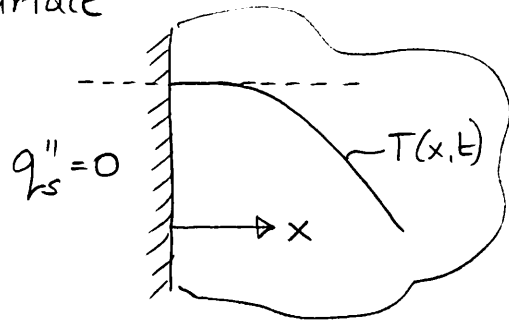
$$-k \left. \frac{\partial T}{\partial x} \right|_{x=0} = q''_s$$

or



(b) Adiabatic or Insulated Surface

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0$$



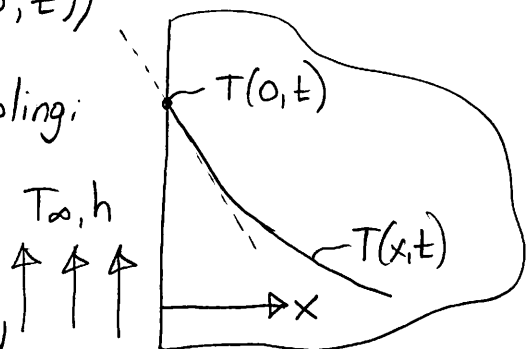
③ Robin (3'rd kind) or Convection surface condition

$$-k \left. \frac{\partial T}{\partial x} \right|_{x=0} = h(T_\infty - T(0, t))$$

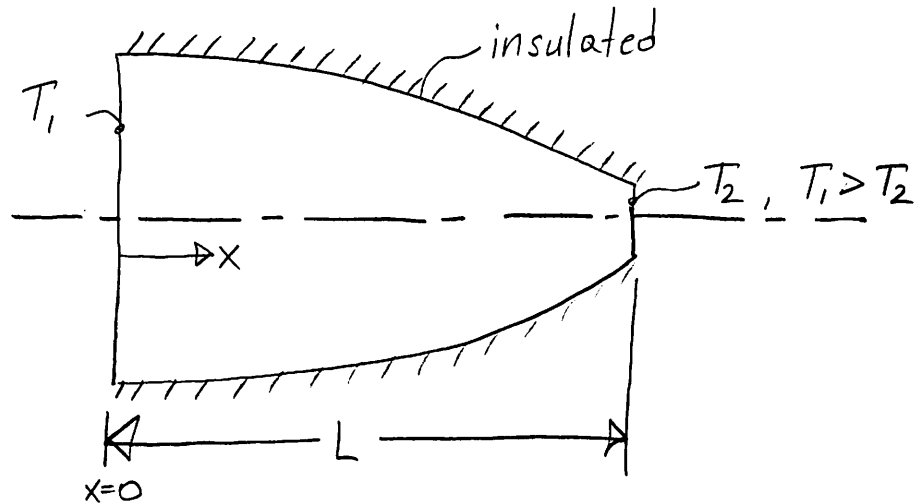
Stems from Newton's Law of cooling:

$$q'' = h(T_s - T_\infty)$$

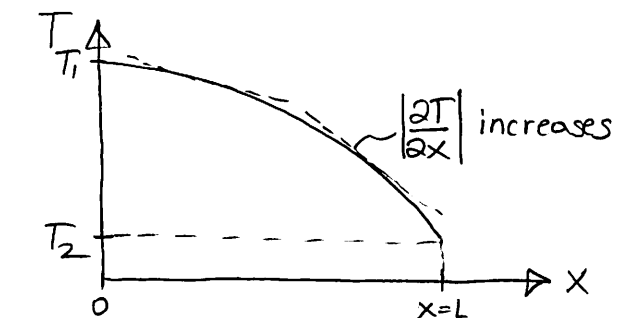
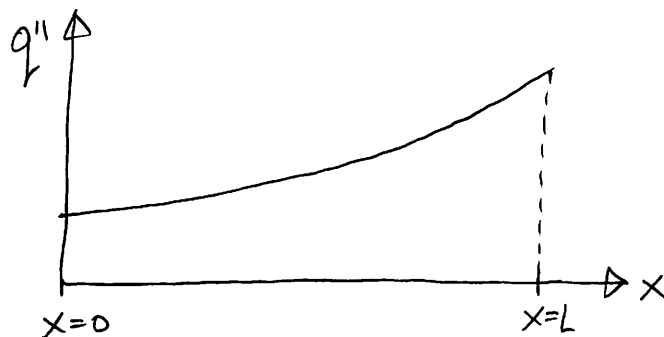
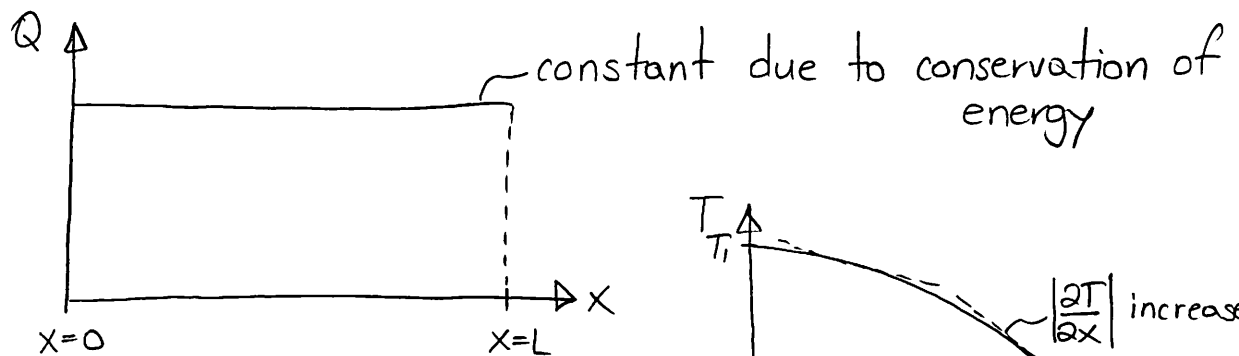
↳ Says heat flux is proportional to some constant h , and temperature difference ΔT .



Example | Assume 1-D, steady-state, heat conduction through the axisymmetric shape below. Assume no heat generation & that it is well insulated on the outsides. Assume constant properties.



Sketch the temperature distribution, heat flux distribution, and heat transfer distribution as a function of x . Sketch only, no need for detailed calculations.



$$\Rightarrow Q = -k A_x \frac{\partial T}{\partial x} = \text{constant}$$

$$A_x \text{ is decreasing so } \frac{\partial T}{\partial x} \uparrow$$

$$q'' = \frac{Q}{A_x} \equiv \text{increasing w/ } x$$

Steady-State 1D Conduction (Chapter 3 of textbook)

So far we've developed the governing equation of heat conduction in a stationary medium, and the associated boundary conditions. Let's apply them to some problems.

Assumptions:

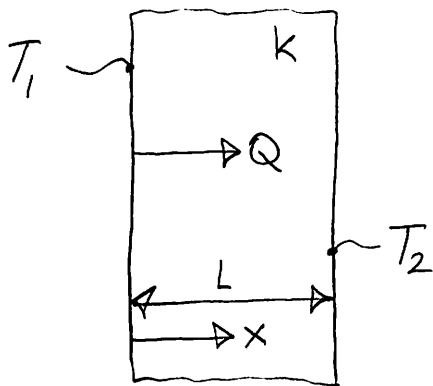
- 1) $\vec{V} = 0 \Rightarrow$ Medium is stationary
- 2) Steady state $\Rightarrow$ not a function of time
- 3) $\dot{Q}''' = 0 \Rightarrow$ no heat generation
- 4) $k = \text{constant} \Rightarrow$ isotropic medium

Let's write down our heat equation:

$$\frac{\partial^2 T}{\partial x^2} + \underbrace{\frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}}_{\text{1-D conduction}} + \underbrace{\frac{\dot{Q}'''}{k}}_{\dot{Q}''' = 0} = \underbrace{\frac{1}{\alpha} \frac{\partial T}{\partial t}}_{\text{Transient term}}$$

So we are left with: $\frac{\partial^2 T}{\partial x^2} = 0 \Rightarrow$ We can solve this easily

① Slab



Note; $Q \equiv$ heat transfer rate [W]

We already know our heat equation governs the heat transfer in the slab, so let's solve it.

$$\frac{\partial^2 T}{\partial x^2} = 0 \quad (\text{integrate once})$$

$$\int \frac{\partial^2 T}{\partial x^2} dx = \int 0 dx$$

$$\frac{\partial T}{\partial x} = C_1 \quad (\text{Integrate once more})$$

$$\int \frac{\partial T}{\partial x} dx = \int C_1 dx$$

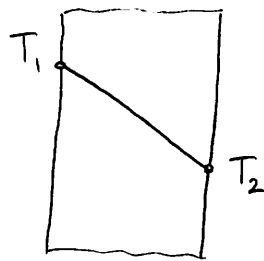
$T = C_1 x + C_2 \Rightarrow$ Now we apply our boundary conditions:

$$T|_{x=0} = T_1 = C_1(0) + C_2 \Rightarrow \boxed{C_2 = T_1}$$

$$T|_{x=L} = T_2 = C_1 L + C_2 = C_1 L + T_1 \Rightarrow \boxed{C_1 = \frac{T_2 - T_1}{L}}$$

$$\boxed{T = \frac{T_2 - T_1}{L} x + T_1}$$

$\hookrightarrow$ Temperature profile in the solid: linear and decreasing from hot to cold.

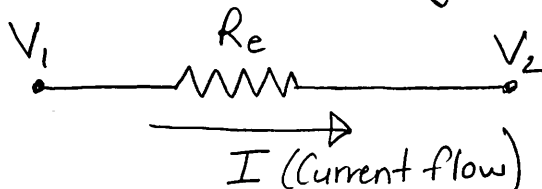


Heat transfer: Fourier's Law

$$Q = -kA \frac{\partial T}{\partial x} = -kA \left(\frac{T_2 - T_1}{L} \right)$$

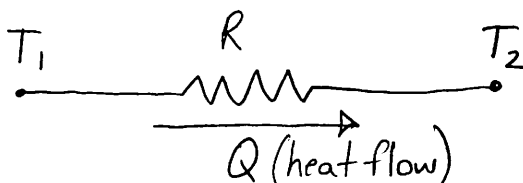
$$\text{Heat flux: } q'' = -k \left(\frac{T_2 - T_1}{L} \right)$$

Note here we can see an interesting analogy.
In electrical engineering:



$$I = \frac{V_1 - V_2}{R_e} = \frac{\overset{\substack{\text{Voltage differen.} \\ \downarrow}}{\Delta V}}{\underset{\substack{\uparrow \\ \text{Electrical} \\ \text{resistance}}}{R_e}}$$

Can we do the same for heat transfer:



$$Q = \frac{T_1 - T_2}{R} = \frac{\Delta T}{R}$$